

Quantum Chemical Computation of Vibrational Spectra for CN-PAHs

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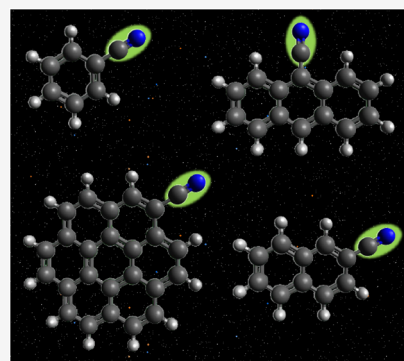
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ABSTRACT: The long theorized polycyclic aromatic hydrocarbons (PAHs) crashed onto the astrochemical scene with the radioastronomical discovery of cyanobenzene (benzonitrile) in 2018, but the astronomical observation of the CN stretch for this or any other known CN-PAHs in the IR has yet to be reported. With the wealth of James Webb Space Telescope (JWST) data being actively returned, the need for reference data for IR features of known interstellar molecules like CN-PAHs has never been greater. Due to complexities in working with PAHs in the laboratory, quantum chemistry provides the best means of high-throughput vibrational reference data generation. This work gives an overview for the state-of-the-art in quantum chemical approaches and results for computing IR spectroscopic data for CN-PAHs, specifically for the CN stretch. The use of the hybrid rDSD/junTZ + B3LYP/N07D method through quartic force fields (QFFs) and second-order vibrational perturbation theory (VPT2) has computed the CN stretch of C_{2v} 9-cyanoanthracene to be 2207 cm^{-1} ($4.531\text{ }\mu\text{m}$), exactly the same as free electron laser experiments report. Additionally, due to the structural regularity of the attached PAHs, the CN stretch does not vary greatly (less than 20 cm^{-1}) from CN-PAH molecule to different CN-PAH molecule. However, this spectroscopic region is home to Mg IV and Ar VI lines that may hinder observations of the CN stretch in astronomical environments where vibrational emission is possible.

KEYWORDS: computational chemistry, IR spectroscopy, astrochemistry, aromatic compounds



1. INTRODUCTION

In the 1980s interstellar IR emission features observed by the first generations of space and airborne telescopes were hypothesized to be linked to polycyclic aromatic hydrocarbons (PAHs).^{1–5} Finally, after much debate as to the reality of such a claim for over 30 years, the first PAH-type molecule was conclusively observed in the interstellar medium in 2018. This came in the form of a benzene derivative, namely benzonitrile with its cyano (CN) group.⁶ While some clues as to benzene's presence had been previously noted with IR observations,⁷ rotational spectroscopy and the ladder of lines that this molecular behavior creates has long-been the workhorse of astronomical molecular detection. For this particular detection, the CN group creates a massive permanent electric dipole moment allowing cyanobenzene/benzonitrile ($CN-C_6H_5$), the polar benzene derivative, to be observed conclusively. This paved the way for subsequent observations of larger so-called "CN-PAHs" including 1- and 2-cyanonaphthalene ($CN-C_{10}H_7$),⁸ 2-cyanoindene ($CN-C_9H_7$),⁹ all three isomers of cyanopyrene ($CN-C_{16}H_9$),^{10,11} and cyanocoronene ($CN-C_{24}H_{11}$).¹² Additionally, some pure PAHs are now known in the interstellar medium (ISM), as well, including indene,^{13,14} benzyne,¹⁵ and, recently, phenalene¹⁶ in addition to James Webb Space Telescope (JWST) observations in the IR confirming benzene in a stellar disk.¹⁷ Hence, PAHs are now conclusively known in space, and the CN-PAHs appear to be the best tracers for them with current technologies.

However, in the age of JWST, more IR data are streaming back to Earth faster than can rationally be interpreted. Reference molecular vibrational spectroscopic data are required and must be produced for proper interpretation of the observations. While the needed rotational spectroscopic data for ground-based observation have been determined for these observed CN-PAHs, IR spectroscopic data for the functionalized PAHs are not as readily available for comparison to JWST observations. Isolated PAHs of any kind are notoriously difficult to isolate in the laboratory for any spectroscopic technique. Their π electrons promote non-covalent interactions making them stick together. Additionally, their similar structures and resulting, related vibrational behavior make separating and uniquely assigning IR molecular features exceedingly difficult to do. While high-resolution, gas phase IR spectra of a few small, pure PAHs are known,^{18–21} no complete IR spectroscopic data have been experimentally produced for CN-PAHs. In this case, gas phase laboratory spectra are highly coveted to continue benchmarking the computations, as the computations are performed on a single

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molecule at 0 K. Of course, any high quality laboratory spectra, regardless of the method (e.g., matrix isolation with noble gas atoms, p -H₂, etc.), go a long way to determining the accuracy of the computed spectra. In light of these difficulties, and with a friendly call for further laboratory investigation, these computed data are vital for extending multimessenger observations of CN-PAHs into the IR where novel astrophysics can be gleaned from the populations of these known interstellar molecules.

2. QUANTUM CHEMISTRY

2.1. Background. In order to assign an observational or even experimental spectrum, the spectral features must be known. This self-consistent paradox is rectified through quantum chemical predictions. These provide informed predictions as to roughly where the spectral features should be in order to narrow significantly the window of possible assignment. Unlike experimental inferences like spectroscopic signatures which require the practitioner to work out the molecule from the signal, quantum chemistry begins with the molecule and predicts molecular properties such as fundamental vibrational frequencies. This highly useful bottom-up approach is dependent upon two factors, the accuracy of the predicted potential energy surface (PES) and the method employed that interprets the resulting vibrational levels from the PES. However, the employed methods must be accurate enough for the results to be meaningful.

The PES is a function of how the energy of a molecule changes in response to displacement of atomic positions, most often in the form of a normal coordinate. The normal coordinates are the atomic motions of the fundamental vibrational frequencies, and there are $3N - 6$ normal coordinates where N is the number of atoms in the molecule ($3N - 5$ coordinates in linear molecules). The PES for a bond length, for instance, produces a Morse function, and the PES for bond angles and torsions result in periodic functions, albeit some with exceedingly high barriers.²² While quantum chemical computations at points along the coordinate, called “scans,” can produce the total PES for a given coordinate from zero to the asymptote for a stretch as an example, such an approach is intractable for all but diatomic molecules. The total number of points that would be required for a scan of each coordinate would be massive enough in sampling the coordinate from a start of roughly half the optimized bond length to some approach to the dissociation limit with meaningful step sizes on the order of 0.1 Å. Then, the global PES would require energy computations for all of the combinations of all coordinates for all intersections of points. Such a construction results in geometric scaling on top of geometric scaling.

As a result, most vibrational frequencies are computed harmonically. These are relatively simple approximations that model each PES as a parabola and originate from Hooke’s law and the typical quantum mechanical harmonic oscillator. These are decent approximations for regions around equilibrium, but even the first fundamental frequency of a given normal coordinate within the harmonic approximation is known to produce errors, typically predicting frequencies that are too high. However, these are relatively inexpensive computations of the PES and get close enough for most applications such as determining the vibrational zero-point energy contribution to the molecular partition function. Scaling factors for reducing the frequency closer to

experimental values have been employed for decades, and these have opened quantum chemical computations to the predictions of PAH spectra even up to hundreds of carbon atoms in size.^{23–26} However, harmonic vibrational frequencies can miss or over/underestimate vital regions of the spectrum even when scaled, especially when two quanta frequencies (overtones and combination bands) are needed to interpret an IR spectrum of PAHs. Consequently, anharmonicity is required.

Simply, anharmonicity is any quantum chemical model of a vibrational PES that is beyond harmonic. The most economic version of such a potential is the quartic force field (QFF), a fourth-order Taylor series expansion of the internuclear Watson Hamiltonian.^{27,28} It takes the mathematical form of

$$V = \frac{1}{2} \sum_{ij} F_{ij} \Delta_i \Delta_j + \frac{1}{6} \sum_{ijk} F_{ijk} \Delta_i \Delta_j \Delta_k + \frac{1}{24} \sum_{ijkl} F_{ijkl} \Delta_i \Delta_j \Delta_k \Delta_l \quad (1)$$

The force constants are represented by the $F_{ij\dots}$ terms for the displacements, $\Delta_i \Delta_j \dots$, of coordinates i, j , etc. The reference for this PES is an optimized geometry at the same or nearly the same level of theory since the QFF must be at the minimum of the PES rendering the first derivative to be zero.^{29–31}

The harmonic approximation is a quadratic Taylor series, and the cubic Taylor series does not have proper limiting behavior, e.g., the potential turns over and goes to negative infinity. While a QFF goes to positive infinity, it has more of the Morse-like shape for bond lengths and closely mirrors the truer periodic functions of the angle/torsional coordinates.²² Hence, the QFF is really the smallest approximation to the anharmonic potential possible. The QFF definition of the PES is most often paired with second-order vibrational perturbation theory (VPT2) which builds upon the harmonic approximation for refining the predicted vibrational levels.^{32–37} This combination of QFFs with VPT2 is often sufficient to achieve accuracies in the prediction of fundamental vibrational frequencies to within 1.0 cm^{−1} of experiment on multiple occasions^{38–47} and most often to within 6.0 cm^{−1}.^{48–50} However, this accuracy is largely dependent upon the choice of the underlying electronic structure method.

These most accurate QFF VPT2 anharmonic frequencies have relied upon coupled cluster theory at the singles, doubles, and perturbative triples level [CCSD(T)]^{51–53} and recently with the addition of explicit correlation (F12)^{54,55} for faster convergence to the complete basis set (CBS) limit.⁵⁶ The energies defining these accurate QFFs are often composite in nature with corrections for core electron correlation, relativity, and even higher-level correlation,^{38–40,57} but these composite approaches begin to break down for large amplitude motions.^{58–60} Even without this shortcoming of accurate composite methods, the largest QFFs have been limited to molecules with less than ten atoms as the computational costs are often much too great for molecules of this size to be treated with existing computing hardware.^{61–63} Despite the relatively sparse nature of QFFs, they still scale geometrically with the number of atoms in the molecule. As such, the largest coupled cluster QFF ever undertaken has actually been for cyanobenzene,⁶⁴ again the first CN-PAH class molecule observed in the ISM.

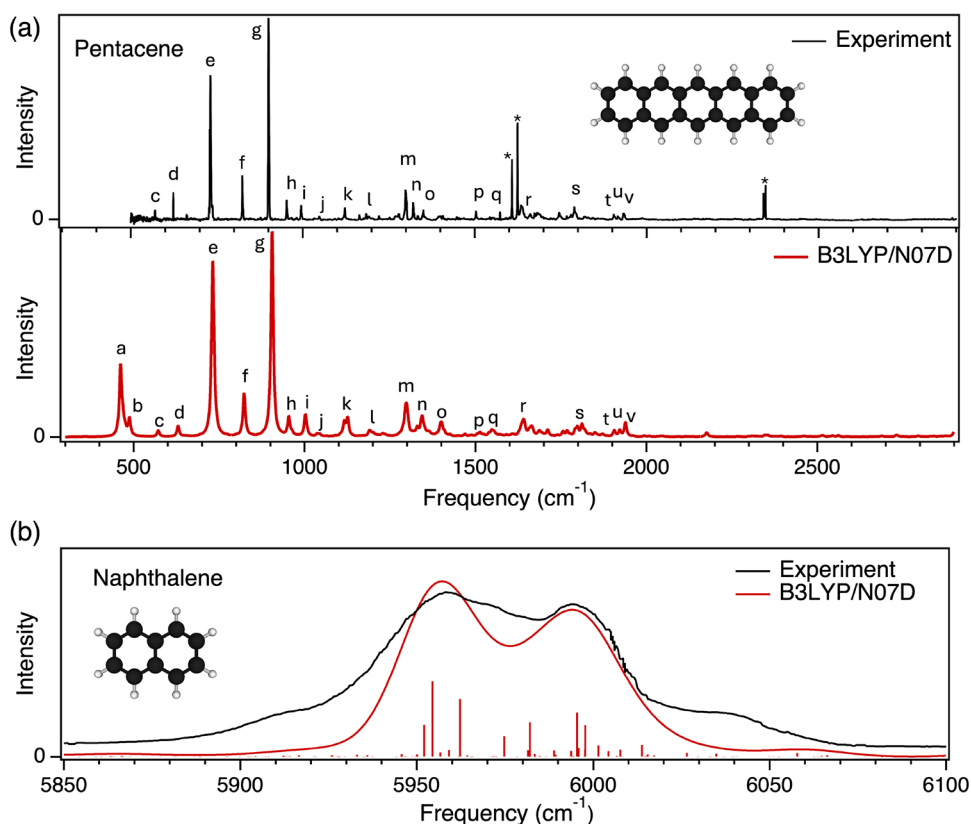


Figure 1. PAH IR absorption spectroscopic benchmarking studies using the B3LYP/N07D level of theory for the anharmonic frequency computations of (a) pentacene in the 500–2000 cm^{-1} with each assigned transition marked with a letter (asterisks denote water contamination in the experimental spectrum),⁷³ and (b) naphthalene in the CH stretch overtone region (5850–6100 cm^{-1}).⁷⁴ Panel (a) reprinted from Anharmonic IR Absorption Spectra of the Prototypical Interstellar PAHs Phenanthrene, Pyrene, and Pentacene in Their Neutral and Cation States, Esposito, V. J., Allamandola, L. J., Boersma, C., Bregman, J. D., Fortenberry, R. C., Maragkoudakis, A., Temi, P., *Molecular Physics*, 2023, by permission of the publisher (Taylor & Francis Ltd., <http://www.tandfonline.com>).

The CCSD(T)-F12/cc-pVDZ-F12 QFF for cyanobenzene requires 21710 normal coordinate, semidiagonal points for this C_{2v} molecule. This approach is still notably accurate producing fundamental vibrational frequencies in line with existing benchmarks.⁴⁹ The CN stretching fundamental, for instance, is computed at this level to be 2238 cm^{-1} ,⁶⁴ exceptionally close to the gas phase experimental value of 2232 cm^{-1} .^{65,66} Utilizing coarse-grained parallelization with distribution across multiple compute nodes, this QFF still required many months of wall time to complete due to the computational cost of each individual coupled cluster job. Hence, functionalized benzene may be the largest coupled cluster QFF computed for some time since even a step up in PAH size to naphthalene requires 104580 normal coordinate, semidiagonal points for this D_{2h} molecule, a nearly 5-fold increase in the number of points. Each individual energy would also roughly double the computational cost for each point. Hence, a coupled cluster QFF for cyanonaphthalene is an order of magnitude more computationally costly endeavor than cyanobenzene. Coupled cluster theory, as accurate as it can be, has just about been maxed out for anharmonic frequency predictions. However, the much less costly density functional theory (DFT) electronic structure approach has shown that it can be applied for the accurate computation of anharmonic vibrational spectroscopic data for PAHs. While molecular dynamics, machine learning, and combinations thereof have recently been leveraged for computing PAH spectra,⁶⁷ we will limit our

discussion to the current state of more traditional quantum chemical approaches.

2.2. DFT QFFs. The use of DFT for QFFs is not new. It is not even new for PAHs. Such computations have been reported for nearly a decade now. In fact, the most detailed IR spectra of PAHs were greatly informed by such computations.^{19–21} The growth of computational hardware as well as efficient and parallel processing has always led to an increase in what quantum chemistry can provide to the larger chemical community, and we have finally arrived at a time where truly anharmonic frequencies can be computed for molecules the sizes of PAHs. Of course, the software had to accompany the hardware, and mathematical tricks can be employed to reduce the computational cost of the QFF. The availability of analytic Hessians (effectively second-derivatives) in Gaussian⁶⁸ and other quantum chemical programs^{69,70} has been around since the 1980s.^{71,72} The use of analytic gradients reduces the computational cost for harmonic frequencies in a harmonic force field since individual energies making up the numerical potential do not have to be computed; equations of the first- and second-derivatives are on hand to provide these additional data. As a result, anharmonic computations for self-consistent field (SCF) based methods like DFT can utilize analytic Hessians to compute numerical displacements of these Hessians to produce a QFF for roughly the computational scaling of a numerical harmonic force field, significantly reducing the overall cost of the QFF compared to a purely numerical QFF. Higher-order methods like the various

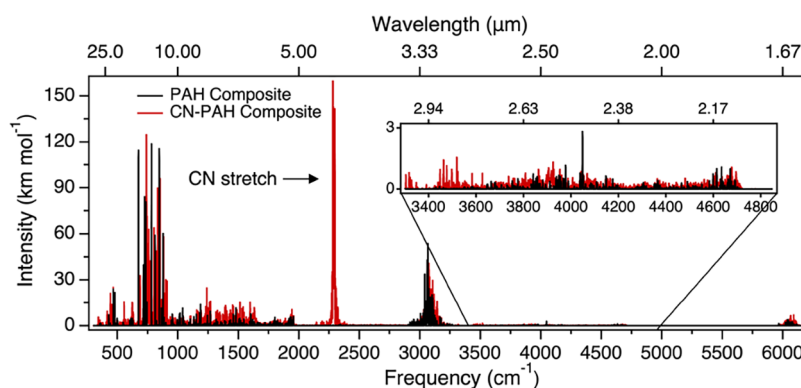


Figure 2. Composite computed anharmonic spectra of cyano-substituted PAHs (red) and standard PAHs (black) from 300 to 6200 cm^{-1} ranging from benzene to pyrene. The inset displays an expanded view of the region from 3350 to 4850 cm^{-1} . Adapted with permission from ref 92. Copyright 2025 American Chemical Society.

truncation levels of coupled cluster theory can also employ analytic Hessians, but these are difficult to derive and implement. Few, if any, quantum chemistry programs today have analytic Hessians available for the most accurate methods such as coupled cluster theory with most Hessians available for methods based on SCF-type procedures. DFT methods are typically SCF-based in some fashion.

Initial studies employed the B97–1 functional with the TZ2P basis set in an attempt to pair a functional that balances basis set size where the spectrum could be treated sufficiently but would not be too costly.^{19,20,75–77} However, the TZ2P basis set was too large for moving to PAHs of the size believed to be present in the ISM, and B97–1/TZ2P had begun to show instability in its treatment of the QFF.⁷⁸ Additionally, the research group led by Vincenzo Barone based in Pisa had demonstrated shortly before this that B3LYP^{79–81} with the N07D basis set was actually well-constructed to deal with anharmonic vibrational frequency predictions.^{82,83} This stand-out hybrid functional has been applied for nearly every chemistry problem since its inception almost 40 years ago, and this behavior has been born out in subsequent PAH benchmarking studies^{21,73,74,78,84–87} as shown in Figure 1. The N07D basis set is effectively the small 6–31G basis set produced in John Pople's group in the 1970s,⁸⁸ but additional functions have been added and optimized by the Pisa group to contain one additional *p* function for H and one each additional *s*, *p*, and *d* function for C and N.⁸⁹ These additional functions appear to be more than sufficient for computing anharmonic frequencies with DFT. Hence, B3LYP/N07D has emerged as the level of theory of choice for the computation of anharmonic vibrational frequencies of PAHs and their functionalized counterparts.

Beyond the approach to defining the QFF and the resulting anharmonic frequencies, the next-biggest need with regards to computing anharmonic DFT vibrational spectroscopic data are related to the intensities. Fortunately, the Pisa group has also provided anharmonic means of computing intensities^{90,91} in addition to anharmonic frequencies as implemented in the Gaussian16 program.⁶⁸ The analytic Hessians are once again useful as the intensities are derivatives of the permanent electric dipole moment in response to the change in nuclear positions, and the reduced cost of the Hessians with DFT makes computation of the anharmonic intensities as well as the frequencies much more efficient. As such, Gaussian16 out of the box (with the addition of the extra N07D terms in the basis

set) is well-equipped to produce anharmonic frequencies of PAHs. A reasonable assumption would then be that anharmonic frequencies of CN-PAHs should be just as straightforward as their pure hydrocarbon parent molecules.

3. CN-PAHS

Initial computations of CN-PAH anharmonic vibrational spectra appeared to behave with the same accuracy as pure PAHs, and they show that the CN stretches should dominate IR spectra of such molecules.⁹³ The majority of the hydrocarbon skeleton features are aligned between pure and CN-PAHs as shown in Figure 2,⁹² even if the CN-PAHs often exhibit more IR features mostly as a result of decreased symmetry. For instance naphthalene drops from D_{2h} symmetry to C_s for its two cyanated isomers, opening many more frequencies to exhibit nonzero intensity. The increase in the permanent electric dipole moment resulting from the addition of the exceptionally electron dense $\text{C} \equiv \text{N}$ group also increases the intensities of the transitions more so than simply reducing the symmetry alone. As such, the spectra of any given region are denser for the CN-PAHs, but the gross placements of the frequencies are still closely aligned for PAHs whether they are pure or CN-functionalized.

However, the dominant feature of a CN-PAH is undoubtedly the CN stretch as shown in Figure 2. The intensities of this region are far greater than any of the other atomic vibrations as is expected from the triple bond and lone pair present in this group. The other characteristic change from PAHs to CN-PAHs is that the CN stretching fundamental itself does not shift greatly as the PAH size or structure changes. From CN-PAHs with a benzene core to a pyrene core, the CN stretch stays within the same 20 cm^{-1} window from 2280 cm^{-1} to 2300 cm^{-1} as computed by B3LYP/N07D.⁹³ CN stretches in small molecules can range from below 1800 cm^{-1} to nearly 2300 cm^{-1} as a result of the surrounding bonding environment and molecule.⁹⁴ However, the relatively consistent nature of PAHs, regardless of size, appears to keep the CN stretch in basically the same range. The isomers of a given CN-PAH exhibit the notably intense CN stretching fundamental across this same 20 cm^{-1} range (those of the three isomers of cyanopyrene occur at 2298 cm^{-1} , 2283 cm^{-1} , and 2288 cm^{-1} for example), and this is true of the entire family of CN-PAHs,⁹³ an exceptionally useful result for classifying CN-PAH astronomical IR spectra. In spite of all of these promising results, the problem with these

B3LYP/N07D CN stretching frequencies is that they are wrong.

Thankfully, only the CN stretch appears to be wrong. Experiment actually places the CN stretch of cyanobenzene at 2232 cm^{-1} .⁶⁵ This is 49 cm^{-1} below the B3LYP/N07D predicted fundamental.⁹³ Again, the massively computationally costly CCSD(T)-F12b/cc-pVDZ-F12 QFF on benzonitrile computes the CN stretch to be 2238 cm^{-1} which is much closer to experiment,⁹³ as it should for such a level of electron correlation. Therefore, the QFF approach and VPT2 do not appear to be the issue if a good level of theory is employed to compute the PES, but coupled cluster theory is, once more, impractical for molecules the size of PAHs or related CN-PAHs. These data points suggest that B3LYP cannot properly treat the rich electron correlation present in the electron dense $\text{C}\equiv\text{N}$ bond. Hence, new functionals are needed that can address this issue but not increase the computational cost for computing the vibrational spectroscopic data of CN-PAHs. Fortunately, work by Malgorzata Biczysko, a member of the same Pisa group, has provided a means of addressing this shortcoming through hybrid methods.⁹⁵

Hybrid methods, in this usage of the term for QFFs, compute the harmonic and anharmonic derivatives at different levels of theory. The harmonic approximation contributes roughly 95% or more of the total value to any frequency but requires often less than 5% of the total QFF computational cost.⁹⁶ This means the harmonics can be computed at a higher level. The anharmonic portions, which are the largest computational cost but only provide a small correction, can be computed at a lower level.⁵⁰ Of course, the anharmonic corrections are essential and should not be neglected, as they are necessary for predicting overtone and combination bands correctly among other things. However, even an egregious 20% error for a lower level method that contributes 5% of the total anharmonic frequency correction results in a decent prediction of the vibrational spectroscopic data with just a 1% total error. Better methods can reduce the error even more while still keeping fast QFFs a possibility. This is different from simply adding the lower-level anharmonic correction frequency to the higher-level anharmonic frequency as the force constants are combined giving an actual QFF instead of just frequencies akin to the difference between CCSDT and CCSD(T). Constructing the full QFF from these different levels provides for the ability to treat resonances in the VPT2 procedure^{50,97} and can open such methods for use in vibrational emission predictions.⁹⁸

In any case, Xu and Biczysko⁹⁵ propose a method that computes harmonic frequencies with the double-hybrid DSDPBEP86 (rDSD) functional and the jun-cc-pVTZ basis set combined with cubic and quartic terms from B3LYP and the SNSD basis set.^{83,99,100} This approach predicts the CN stretch of cyanobenzene to be 2227 cm^{-1} ,⁹⁵ exceptionally close to the experimental value, again, of 2232 cm^{-1} . Now, if the entire QFF (harmonics and anharmonics) are computed with rDSD/jun-cc-pVTZ alone, the harmonics are actually too low in frequency. This implies that any anharmonic corrections within a pure rDSD QFF would not be correct, either. Combining these unphysically low, rDSD harmonic force constants with the B3LYP anharmonics appears to situate the frequency in just the right spot.^{64,95} While the double-hybrid is nominally more computationally costly than B3LYP ($O(N^5)$ versus $O(N^4)$, respectively), this is still two orders of

polynomial scaling less than the single, noniterative (T) portion of CCSD(T) which is $O(N^7)$, where N is the number of basis functions. While the jun-cc-pVTZ basis set is larger (i.e., N is bigger) than the cc-pVDZ-F12 basis set used with coupled cluster theory for cyanobenzene,⁶⁴ the scaling of the rDSD method more than makes up for this growth in basis set size. Hence, the hybrid, so-called “rDSD/junTZ + B3LYP/SNSD” QFF is capable of computing anharmonic frequencies for larger CN-PAHs, and employing such a hybrid approach for computing anharmonic frequencies is available in the Gaussian16 program with a simple keyword (Freq = ReadDifferenharmonic).¹⁰¹

Followup work has shown that the rDSD/junTZ + B3LYP/SNSD hybrid QFF and the related rDSD/junTZ + B3LYP/N07D QFF produce nearly identical CN stretches in cyanobenzene of 2227 cm^{-1} and 2226 cm^{-1} , respectively. Both are less than 6 cm^{-1} below experiment.⁶⁴ Hence, the change in the basis set for the lower-cost anharmonic portion does not greatly affect the outcome. Beyond cyanobenzene, novel free-electron laser experiments at FELIX in Nijmegen, Netherlands have measured a CN stretch frequency for C_{2v} 9-cyanoanthracene at 2207 cm^{-1} . The rDSD/junTZ + B3LYP/N07D hybrid QFF value is also 2207 cm^{-1} .⁶⁴ While the pure B3LYP/N07D QFFs are unable to accurately predict the absolute location of the CN stretches in CN-PAHs, the narrow behavior of the CN stretch still appears to be correct even from the experimental values of cyanobenzene and 9-cyanoanthracene. Even so, more accurate QFF computational vibrational spectroscopic data are needed for a larger library of CN-PAHs in order to provide the needed reference data for JWST observations and for characterizing the vibrational behavior of CN-PAHs.

4. ASTROPHYSICAL IMPLICATIONS

However, astronomically observing these IR stretches of the CN group may not be straightforward. First, the known CN-PAHs have been found in cold molecular clouds like TMC-1. Such regions can emit in the radio since most any area of space possesses enough ambient energy to excite molecular rotations. This is not the case in IR spectroscopic ranges that require higher-energy excitations. Hence, the phenomena within astronomical environments that make PAHs visible in other astronomical environments are not at work in places where the CN-PAHs are currently known. The CN stretch may not emit in such regions. The abundance of such CN-PAHs in regions that exhibit the aromatic infrared bands (AIBs), and thus would be suitable for the detection of the CN stretch emission, will depend heavily on their photostability and formation/destruction mechanisms, each of which is currently being investigated by the community.

Next, the CN stretch has been hypothesized to potentially interfere with the CD stretch in deuterated PAHs.¹⁰² While that may be the case to some extent, the difference in spectral shape should be sufficient for the CN stretch to stand out. Notably, the CN stretch for most PAH sizes is, again, known to fit within a 20 cm^{-1} ($0.03\text{ }\mu\text{m}$) window around 2220 cm^{-1} ($\sim 4.5\text{ }\mu\text{m}$). The CD stretch also falls in this window, but it produces broader features and extends far beyond the narrow width of the CN stretch of CN-PAHs from roughly 2080 cm^{-1} to 2270 cm^{-1} (4.4 to $4.8\text{ }\mu\text{m}$).¹⁰³ As such, the narrow FWHM of the CN stretch and large intensity of this fundamental

should make a sharp feature for clear identification even when CD stretches are present.

However, spectral features besides the CD stretch are present in this spectroscopic region, as well. Specifically, Mg IV at 2230 cm^{-1} ($4.48\text{ }\mu\text{m}$) and Ar VI at 2208 cm^{-1} ($4.53\text{ }\mu\text{m}$) will also produce very narrow lines directly in interference with the CN stretch. Granted a “narrow” molecular feature such as the CN stretch, especially over the 20 cm^{-1} window, will always be wider than most any atomic line as observed with the same instrument (i.e., NIRSpec onboard JWST in this case). Hence, shoulders and deviations of the Mg IV and Ar VI features may belie the presence of CN-PAHs, but the unique widening from the CN stretch of CN-PAHs within these atomic lines may still be difficult to see. Ambient temperatures in the ISM (e.g., 100 K) contribute to rotational temperature equilibrium that impacts the rotational envelope of a rovibrational band but does not impact the band center (i.e., frequency). However, internal vibrational temperature has an effect on band position where temperature shifts within the IR spectrum of PAHs are expected to be small (on the order of $-0.02\text{ cm}^{-1}\text{ K}^{-1}$ in the CH region.) In “hot” UV photoexcited PAHs ($T_{\text{vib}} \sim 1000\text{ K}$), this might induce a red shift of up to 20 cm^{-1} , although the shift is expected to be smaller for larger PAHs that have a higher density of vibrational states available for the intramolecular vibrational energy redistribution prior to IR emission. Even so, modern instruments and emerging analytical algorithms may yet help showcase the CN stretch uniquely.

Finally, the CN (or NC) stretches of the isocyanoPAHs (NC-PAHs) have also been predicted with B3LYP/N07D once more. These computations show that the NC stretches fall within a narrow window itself, but they lie roughly 100 cm^{-1} to 150 cm^{-1} below that of the CN stretch in CN-PAHs. The NC stretches in NC-PAHs are also more intense than their CN-PAH cousins which would make the NC-PAHs potentially observable in regions where CN-PAHs could also be observed in the IR.¹⁰⁴ Formation of NC-PAHs would likely need to precede CN-PAHs as the former can be turned into the latter net exothermically as long as there is an energy source, but the net exothermicity precludes the return reaction.¹⁰⁵ Even so, NC-PAHs are stable and should have some appreciable lifetime. Granted, rDSD/junTZ + B3LYP/N07D QFFs should be performed on the NC-PAHs, as well, to verify the positions of the NC stretches, but the benchmarks from the shifts in the structure and hybrid QFF treatment of the CN-PAHs place these NC stretches in the range of 2100 cm^{-1} ($\sim 4.7\text{ }\mu\text{m}$).

5. CONCLUSIONS

CN-PAHs may have been hiding in plain sight within astronomical IR spectra since such observations began. While the CN stretch in CN-PAHs is somewhat surprisingly narrow across various shapes and sizes of PAHs and has a large intensity, astrophysical conditions, coincidence with atomic lines, and, likely to a smaller extent, placement of CD stretches may be hiding the CN stretches within PAHs from observations. Beyond this, the non-CN-based molecular vibrations of CN-PAHs are at similar frequencies as pure PAHs, but their likely lower abundance will hide even those features different from their pure PAH counterparts. The rotational detections of CN-PAHs are certain, but their IR observations may yet take time. However, the data produced through quantum chemical computations and backed up by

state-of-the-art experiments should be useful markers for those willing to search IR observations where CN-PAHs may emit in the IR. Such data can and have been included in large PAH spectral databases where the features for these various molecules can be mixed together showcasing the overall trends that PAHs produce in IR observations.^{23,106–109}

Future CN-PAH vibrational spectroscopic data for comparing to such astronomical IR observations can benefit from the use of hybrid QFFs such as rDSD/junTZ + B3LYP/SNSD and its closely related rDSD/junTZ + B3LYP/N07D counterpart. Such methods balance accuracy and computational cost while still treating the electron density of the $\text{C}\equiv\text{N}$ region that B3LYP alone struggles to capture. While pure B3LYP/N07D QFFs provide exceptional accuracy for creating simulated IR spectra of unsubstituted PAHs, the CN stretch is in error by dozens of wavenumbers. The double-hybrid rDSD functional has been shown to capture this behavior well when anharmonic treatment from B3LYP is included. Hence, this method will be utilized at least by these authors in computing the anharmonic spectroscopic data of CN-PAHs.

Each molecule and molecular class presents its own challenges. For most of the history of quantum chemistry, PAHs were always much too large for accurate quantum chemical predictions of vibrational frequencies, but that has changed over the past decade. Then, the computational methods that finally worked for pure PAHs struggled with the class-defining CN stretch in CN-PAHs. Now, it appears that such has been worked out. Questions, of course, remain in moving forward: Can such a hybrid QFF be applied to other functionalized PAHs? How large of PAHs or CN-PAHs can the present computational approaches treat? Will the electronic structure of larger CN-PAHs require a further refinement of the computational methods? Can machine learning or molecular dynamics be applied to computing the IR spectra of these systems? What is the result of charge on the CN stretch or CN-PAH IR spectra in general? And many other questions not presently foreseen are also likely to arise. However, this is the nature of scientific exploration and creativity. The exploration of quantum chemistry for astrochemical applications may yet open new doors and will certainly provide for creative solutions both now and in the future. Attempting to address and answer these problems may have impact elsewhere, as PAHs and likely CN-PAHs are exceptionally common here in Earth, especially from industrial sources and really anywhere hydrocarbons are burned. The motivations may be different, but the overlap is certainly more than zero. The exploration of quantum chemistry for astrochemical applications may yet open new doors and will certainly provide for creative solutions both now and in the future.

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Notes

The authors declare no competing financial interest.

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